



IIT ROORKEE



NPTEL ONLINE
CERTIFICATION COURSE

RBD

Dr. A. Ramesh
DEPARTMENT OF MANAGEMENT
IIT ROORKEE



Learning Objectives

- Estimate variance components in an experiment involving random factors
- Understand the blocking principle and how it is used to isolate the effect of nuisance factors
- Design and conduct experiments involving the randomized complete block design

Randomized Block Design

- A completely randomized design (CRD) is useful when the experimental units are homogeneous
- If the experimental units are heterogeneous, **blocking** is often used to form homogeneous groups

Why RBD?

- A problem can arise whenever differences due to extraneous factors (ones not considered in the experiment) cause the MSE term in this ratio to become large.
- In such cases, the F value in equation can become small, signaling no difference among treatment means when in fact such a difference exists.

$$F = \frac{MSTR}{MSE}$$

Randomized block design

- Experimental studies in business often involve experimental units that are highly heterogeneous; as a result, randomized block designs are often employed.
- Blocking in experimental design is similar to stratification in sampling.

Randomized block design

- Its purpose is to control some of the extraneous sources of variation by removing such variation from the MSE term.
- This design tends to provide a better estimate of the true error variance and leads to a more powerful hypothesis test in terms of the ability to detect differences among treatment means.

Air Traffic Controller Stress Test

- A study measuring the fatigue and stress of air traffic controllers resulted in proposals for modification and redesign of the controller's work station
- After consideration of several designs for the work station, three specific alternatives are selected as having the best potential for reducing controller stress
- The key question is: To what extent do the three alternatives differ in terms of their effect on controller stress?



Air Traffic Controller Stress Test

- In a completely randomized design, a random sample of controllers would be assigned to each work station alternative.
- However, controllers are believed to differ substantially in their ability to handle stressful situations.
- What is high stress to one controller might be only moderate or even low stress to another.
- Hence, when considering the within-group source of variation (MSE), we must realize that this variation includes both random error and error due to individual controller differences.
- In fact, managers expected controller variability to be a major contributor to the MSE term.

A randomized block design for the air traffic controller stress test

Treatments

Blocks

	System A	System B	System C
Controller 1	15	15	18
Controller 2	14	14	14
Controller 3	10	11	15
Controller 4	13	12	17
Controller 5	16	13	16
Controller 6	13	13	13

Solving this example using ANOVA in python

```
In [1]: import pandas as pd
import numpy as np
import scipy
import statsmodels.api as sm
from statsmodels.formula.api import ols
```

```
In [4]: df = pd.read_excel('RBD.xlsx')
df
```

Out[4]:

	System A	System B	System C
0	15	15	18
1	14	14	14
2	10	11	15
3	13	12	17
4	16	13	16
5	13	13	13

Solving this example using ANOVA in python

ANOVA

```
In [20]: data = pd.melt(df.reset_index(), id_vars=['index'], value_vars=['System A', 'System B', 'System C'])
data.columns = ['index', 'treatments', 'value']
```

```
In [21]: model = ols('value ~ C(treatments)', data=data).fit()
anova_table = sm.stats.anova_lm(model, typ=1)
anova_table
```

Out[21]:

	df	sum_sq	mean_sq	F	PR(>F)
C(treatments)	2.0	21.0	10.500000	3.214286	0.068903
Residual	15.0	49.0	3.266667	NaN	NaN

```
In [9]: # accept the null hypothesis
```

Summary of stress data for the air traffic controller stress test

Treatments → Blocks ↓	System A	System B	System C	Block total	Block means
Controller 1	15	15	18	48	$\bar{x}_{1.} = 16$
Controller 2	14	14	14	42	$\bar{x}_{2.} = 14$
Controller 3	10	11	15	36	$\bar{x}_{3.} = 12$
Controller 4	13	12	17	42	$\bar{x}_{4.} = 14$
Controller 5	16	13	16	45	$\bar{x}_{5.} = 15$
Controller 6	13	13	13	39	$\bar{x}_{6.} = 13$
Column Total	81	78	93	252	$\bar{\bar{X}} = 252/18 = 14$

Summary of stress data for the air traffic controller stress test

- Treatment means

$$\overline{x_{.1}} = 81/6 = 13.5$$

$$\overline{x_{.2}} = 78/6 = 13$$

$$\overline{x_{.3}} = 93/6 = 15.5$$

ANOVA TABLE FOR THE RANDOMIZED BLOCK DESIGN WITH k TREATMENTS AND b BLOCKS

Sources of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F	P- value
Treatments	SS Treatments	$k-1$	MS Treatments = $SSTR/k-1$	MS Treatments / MSE	
Blocks	SS block	$(b-1)$	MSBL = $SSBL/b-1$		
Error	SSE	$(k-1)(b-1)$	MSE = $SSE/(k-1)(b-1)$		
Total	SST	$nt-1$			

RBD Problem

- x_{ij} = value of the observation corresponding to treatment j in block i
- $\bar{x}_{.j}$ = sample mean of the j th treatment
- $\bar{x}_{i.}$ = sample mean for the i th block
- $\bar{\bar{x}}$ = overall sample mean

RBD Problem

Step 1. Compute the total sum of squares (SST).

$$SST = \sum_{i=1}^b \sum_{j=1}^k (x_{ij} - \bar{\bar{x}})^2$$

Step 1. $SST = (15 - 14)^2 + (15 - 14)^2 + (18 - 14)^2 + \dots + (13 - 14)^2 = 70$

Step 2. Compute the sum of squares due to treatments (SSTR).

$$SSTR = b \sum_{j=1}^k (\bar{x}_{.j} - \bar{\bar{x}})^2$$

Step 2. $SSTR = 6[(13.5 - 14)^2 + (13.0 - 14)^2 + (15.5 - 14)^2] = 21$

RBD Problem

Step 3. Compute the sum of squares due to blocks (SSBL).

$$\text{SSBL} = k \sum_{i=1}^b (\bar{x}_{i\cdot} - \bar{\bar{x}})^2$$

Step 3. $\text{SSBL} = 3[(16 - 14)^2 + (14 - 14)^2 + (12 - 14)^2 + (14 - 14)^2 + (15 - 14)^2 + (13 - 14)^2] = 30$

Step 4. Compute the sum of squares due to error (SSE).

$$\text{SSE} = \text{SST} - \text{SSTR} - \text{SSBL}$$

Step 4. $\text{SSE} = 70 - 21 - 30 = 19$

ANOVA table for the air traffic controller stress test

Sources of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F	P- value
Treatments	21	2	10.5	10.5/1.9 =5.53	0.024
Blocks	30	5	6.0		
Error	19	10	1.9		
Total	70	17			

$$F_{.025} = 5.46 \text{ and } F_{.01} = 7.56.$$

Reject the null hypothesis

Solving RBD example using python

```
In [1]: import pandas as pd
import numpy as np
import scipy
import statsmodels.api as sm
from statsmodels.formula.api import ols
```

```
In [4]: df = pd.read_excel('RBD.xlsx')
df
```

Out[4]:

	System A	System B	System C
0	15	15	18
1	14	14	14
2	10	11	15
3	13	12	17
4	16	13	16
5	13	13	13

Solving RBD example using python

```
In [20]: data = pd.melt(df.reset_index(), id_vars=['index'], value_vars=['System A','System B','System C'])
data.columns = ['blocks', 'treatments', 'value']
```

```
In [22]: model = ols('value ~ C(block)+ C(treatments)', data=data).fit()
anova_table = sm.stats.anova_lm(model, typ=1)
anova_table
```

Out[22]:

	df	sum_sq	mean_sq	F	PR(>F)
C(block)	5.0	30.0	6.0	3.157895	0.057399
C(treatments)	2.0	21.0	10.5	5.526316	0.024181
Residual	10.0	19.0	1.9	NaN	NaN

```
In [23]: # reject the null hypothesis
```

Conclusion

- Finally, note that the ANOVA table shown in Table provides an F value to test for treatment effects but *not* for blocks.
- The reason is that the experiment was designed to test a single factor—work station design.
- The blocking based on individual stress differences was conducted to remove such variation from the MSE term.
- However, the study was not designed to test specifically for individual differences in stress.

Problem 2: RBD

- An experiment was performed to determine the effect of four different chemicals on the strength of a fabric.
- These chemicals are used as part of the permanent press finishing process.
- Five fabric samples were selected, and a randomized complete block design was run by testing each chemical type once in random order on each fabric sample.
- The data are shown in Table.
- We will test for differences in means using an ANOVA with $\alpha = 0.01$.

Problem 2: RBD

- Table: Fabric Strength Data—Randomized Complete Block Design

Chemical Type	Fabric Sample					Treatment Totals	Treatment Averages
	1	2	3	4	5	$y_{i\cdot}$	$\bar{y}_{i\cdot}$
1	1.3	1.6	0.5	1.2	1.1	5.7	1.14
2	2.2	2.4	0.4	2.0	1.8	8.8	1.76
3	1.8	1.7	0.6	1.5	1.3	6.9	1.38
4	3.9	4.4	2.0	4.1	3.4	17.8	3.56
Block totals $y_{\cdot j}$	9.2	10.1	3.5	8.8	7.6	39.2($y_{\cdot\cdot}$)	
Block averages $\bar{y}_{\cdot j}$	2.30	2.53	0.88	2.20	1.90	1.96($\bar{y}_{\cdot\cdot}$)	

Anova using jupyter

```
In [3]: df = pd.read_excel('rbd2.xlsx')
df
```

Out[3]:

	chem1	chem2	chem3	chem4
0	1.3	2.2	1.8	3.9
1	1.6	2.4	1.7	4.4
2	0.5	0.4	0.6	2.0
3	1.2	2.0	1.5	4.1
4	1.1	1.8	1.3	3.4

```
In [4]: data = pd.melt(df.reset_index(), id_vars=['index'], value_vars=['chem1', 'chem2', 'chem3', 'chem4'])
data.columns = ['index', 'treatments', 'value']
```

```
In [6]: model = ols('value ~ C(treatments)', data=data).fit()
aov_table = sm.stats.anova_lm(model, typ=1)
aov_table
```

Out[6]:

	df	sum_sq	mean_sq	F	PR(>F)
C(treatments)	3.0	18.044	6.014667	12.589569	0.000176
Residual	16.0	7.644	0.477750	NaN	NaN

Problem 2: RBD

- The sums of squares for the analysis of variance are computed as follows:

$$\begin{aligned}SS_T &= \sum_{i=1}^4 \sum_{j=1}^5 y_{ij}^2 - \frac{y_{..}^2}{ab} \\&= (1.3)^2 + (1.6)^2 + \cdots + (3.4)^2 - \frac{(39.2)^2}{20} = 25.69\end{aligned}$$

$$\begin{aligned}SS_{\text{Treatments}} &= \sum_{i=1}^4 \frac{y_{i.}^2}{b} - \frac{y_{..}^2}{ab} \\&= \frac{(5.7)^2 + (8.8)^2 + (6.9)^2 + (17.8)^2}{5} - \frac{(39.2)^2}{20} = 18.04\end{aligned}$$

Problem 2: RBD

$$\begin{aligned}SS_{\text{Blocks}} &= \sum_{j=1}^5 \frac{y_{.j}^2}{a} - \frac{y_{..}^2}{ab} \\&= \frac{(9.2)^2 + (10.1)^2 + (3.5)^2 + (8.8)^2 + (7.6)^2}{4} - \frac{(39.2)^2}{20} = 6.69 \\SS_E &= SS_T - SS_{\text{Blocks}} - SS_{\text{Treatments}} \\&= 25.69 - 6.69 - 18.04 = 0.96\end{aligned}$$

Problem 2: RBD

- Analysis of Variance for the Randomized Complete Block Experiment

Sources of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F	P- value
Chemical types (Treatments)	18.04	3	6.01	75.13	4.79 E-8
Fabric samples (Blocks)	6.69	4	1.67		
Error	0.96	12	0.08		
Total	25.69	19			

Conclusion

- The ANOVA is summarized in the previous table
- Since $f_0 = 75.13 > f_{0.01,3,12} = 5.95$ (the P -value is 4.79×10^{-8}), we conclude that there is a significant difference in the chemical types so far as their effect on strength is concerned.

Python code for problem 2

```
In [2]: import pandas as pd
import statsmodels.api as sm
from statsmodels.formula.api import ols
from statsmodels.stats.anova import anova_lm
```

```
In [3]: df = pd.read_excel('RBD2.xlsx')
```

```
In [4]: df
```

Out[4]:

	chem1	chem2	chem3	chem4
0	1.3	2.2	1.8	3.9
1	1.6	2.4	1.7	4.4
2	0.5	0.4	0.6	2.0
3	1.2	2.0	1.5	4.1
4	1.1	1.8	1.3	3.4

Python code for problem 2

```
In [7]: data = pd.melt(df.reset_index(), id_vars=['index'], value_vars=['chem1', 'chem2', 'chem3', 'chem4'])
data.columns = ['Fabric samples', 'Chemical types', 'value']
data
```

Out[7]:

	Fabric samples	Chemical types	value
0	0	chem1	1.3
1	1	chem1	1.6
2	2	chem1	0.5
3	3	chem1	1.2
4	4	chem1	1.1
5	0	chem2	2.2
6	1	chem2	2.4
7	2	chem2	0.4
8	3	chem2	2.0
9	4	chem2	1.8
10	0	chem3	1.8
11	1	chem3	1.7
12	2	chem3	0.6
13	3	chem3	1.5
14	4	chem3	1.3
15	0	chem4	3.9
16	1	chem4	4.4
17	2	chem4	2.0
18	3	chem4	4.1
19	4	chem4	2.4

Python code for problem 2

```
In [11]: model = ols('value ~ C(Fabric) + C(Chemical)', data=data).fit()  
         anova_table = sm.stats.anova_lm(model, typ=1)  
         anova_table
```

Out[11]:

	df	sum_sq	mean_sq	F	PR(>F)
C(Fabric)	4.0	6.693	1.673250	21.113565	2.318913e-05
C(Chemical)	3.0	18.044	6.014667	75.894848	4.518310e-08
Residual	12.0	0.951	0.079250	NaN	NaN



IIT ROORKEE



NPTEL ONLINE
CERTIFICATION COURSE

Two Way ANOVA

Dr. A. Ramesh

DEPARTMENT OF MANAGEMENT
IIT ROORKEE



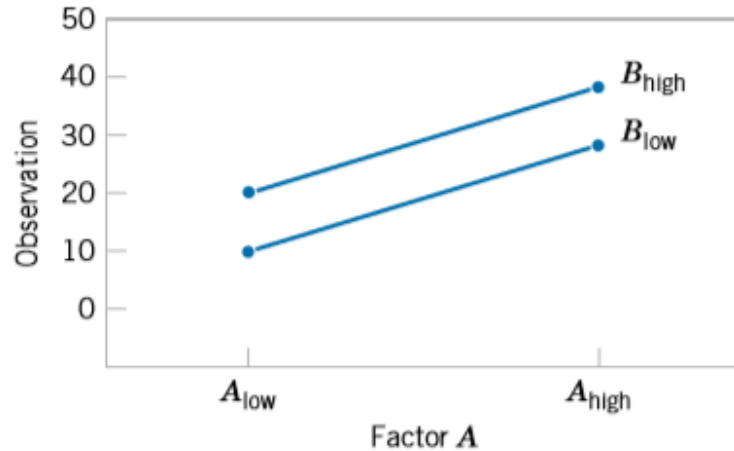
Learning objectives

- Design and conduct engineering experiments involving several factors using the factorial design approach
- Understand how the ANOVA is used to analyze the data from these experiments
- Know how to use the two-level series of factorial designs

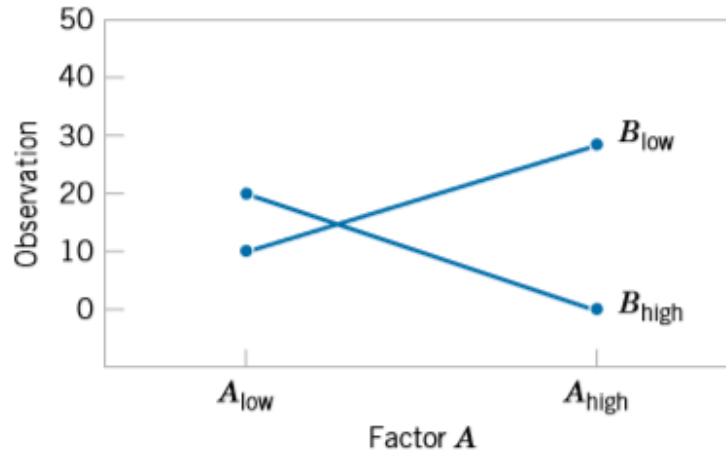
Factorial Experiment

- A **factorial experiment** is an experimental design that allows simultaneous conclusions about two or more factors.
- The term factorial is used because the experimental conditions include all possible combinations of the factors.
- The effect of a factor is defined as the change in response produced by a change in the level of the factor. It is called a main effect because it refers to the primary factors in the study
- For example, for a levels of factor A and b levels of factor B, the experiment will involve collecting data on ab treatment combinations.
- Factorial experiments are the only way to discover interactions between variables.

Factorial Experiment



Factorial Experiment, no interaction



Factorial Experiment, with interaction

Two-factor Factorial Experiments

- The simplest type of factorial experiment involves only two factors, say, A and B.
- There are a levels of factor A and b levels of factor B.
- This two-factor factorial is shown in next table .
- The experiment has n replicates, and each replicate contains all ab treatment combinations.

Two-factor Factorial Experiments

Data Arrangement for a Two-Factor Factorial Design

		Factor <i>B</i>				Totals	Averages
		1	2	...	<i>b</i>		
Factor <i>A</i>	1	$y_{111}, y_{112},$ $\dots, y_{11n}$	$y_{121}, y_{122},$ $\dots, y_{12n}$		$y_{1b1}, y_{1b2},$ $\dots, y_{1bn}$	$y_{1..}$	$\bar{y}_{1..}$
	2	$y_{211}, y_{212},$ $\dots, y_{21n}$	$y_{221}, y_{222},$ $\dots, y_{22n}$		$y_{2b1}, y_{2b2},$ $\dots, y_{2bn}$	$y_{2..}$	$\bar{y}_{2..}$
	$\vdots$						
	<i>a</i>	$y_{a11}, y_{a12},$ $\dots, y_{a1n}$	$y_{a21}, y_{a22},$ $\dots, y_{a2n}$		$y_{ab1}, y_{ab2},$ $\dots, y_{abn}$	$y_{a..}$	$\bar{y}_{a..}$
Totals		$y_{\cdot 1\cdot}$	$y_{\cdot 2\cdot}$		$y_{\cdot b\cdot}$	$y_{\cdot \cdot \cdot}$	
Averages		$\bar{y}_{\cdot 1\cdot}$	$\bar{y}_{\cdot 2\cdot}$		$\bar{y}_{\cdot b\cdot}$		$\bar{y}_{\cdot \cdot \cdot}$

Two-factor Factorial Experiments

- The observation in the ij th cell for the k th replicate is denoted by y_{ijk}
- In performing the experiment, the abn observations would be run in random order.
- Thus, like the single factor experiment, the two-factor factorial is a completely randomized design.

Example

- As an illustration of a two-factor factorial experiment, we will consider a study involving the Common Admission test (CAT), a standardized test used by graduate schools of business to evaluate an applicant's ability to pursue a graduate program in that field.
- Scores on the CAT range from 200 to 800, with higher scores implying higher aptitude.

Three CAT preparation programs.

- In an attempt to improve students' performance on the CAT, a major university is considering offering the following three CAT preparation programs.
 1. A three-hour review session covering the types of questions generally asked on the CAT.
 2. A one-day program covering relevant exam material, along with the taking and grading of a sample exam.
 3. An intensive 10-week course involving the identification of each student's weaknesses and the setting up of individualized programs for improvement.

Factor - 1 , 3 treatment

- One factor in this study is the CAT preparation program, which has three treatments:
 - Three-hour review,
 - One-day program, and
 - 10-week course.
- Before selecting the preparation program to adopt, further study will be conducted to determine how the proposed programs affect CAT scores.

Factor 2 : 3 Treatment

- The CAT is usually taken by students from three colleges:
- the College of Business,
- the College of Engineering, and
- the College of Arts and Sciences.
- Therefore, a second factor of interest in the experiment is whether a student's undergraduate college affects the CAT score.
- This second factor, undergraduate college, also has three treatments:
 - Business,
 - Engineering, and
 - Arts and sciences.

Nine Treatment Combinations for The Two-factor **CAT** Experiment

Factor A: Preparation Program	Factor B: College		
	Business	Engineering	Arts and sciences
Three-hour review	1	2	3
One-day program	4	5	6
10-Week course	7	8	9

Replication

- In experimental design terminology, the sample size of two for each treatment combination indicates that we have two **replications**.

CAT SCORES FOR THE TWO-FACTOR EXPERIMENT

Factor A: Preparation Program	Factor B: College		
	Business	Engineering	Arts and sciences
Three-hour review	500	540	480
	580	460	400
One-day program	460	560	420
	540	620	480
10-Week course	560	600	480
	600	580	410

The analysis of variance computations answers the following questions.

- **Main effect (factor A):** Do the preparation programs differ in terms of effect on CAT scores?
- **Main effect (factor B):** Do the undergraduate colleges differ in terms of effect on CAT scores?
- **Interaction effect (factors A and B):** Do students in some colleges do better on one type of preparation program whereas others do better on a different type of preparation program?

Interaction

- The term **interaction** refers to a new effect that we can now study because we used a factorial experiment.
- If the interaction effect has a significant impact on the CAT scores, we can conclude that the effect of the type of preparation program depends on the undergraduate college.

ANOVA Table for the Two-factor Factorial Experiment with r Replications

Sources of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F	P- value
Factor A	SSA	(a -1)	SSA/a-1	MSA / MSE	
Factor B	SSB	(b-1)	SSB/b-1	MSB/ MSE	
Interaction	SSAB	(a-1)(b-1)	MSAB = SSAB/(a-1)(b-1)	MSAB / MSE	
Error	SSE	ab(r-1)	MSE= SSE/(ab)(r-1)		
Total	SST	n _T -1			

Abbreviation

a = number of levels of factor A

b = number of levels of factor B

r = number of replications

n_T = total number of observations taken in the experiment; $n_T = abr$

ANOVA Procedure

- The ANOVA procedure for the two-factor factorial experiment requires us to partition the sum of squares total (SST) into four groups:
 - sum of squares for factor A (SSA),
 - sum of squares for factor B (SSB),
 - sum of squares for interaction (SSAB), and
 - sum of squares due to error (SSE).
- The formula for this partitioning follows.

$$SST = SSA + SSB + SSAB + SSE$$

Computations and Conclusions

x_{ijk} = observation corresponding to the k th replicate taken from treatment i of factor A and treatment j of factor B

$\bar{x}_{i\cdot}$ = sample mean for the observations in treatment i (factor A)

$\bar{x}_{\cdot j}$ = sample mean for the observations in treatment j (factor B)

$\bar{x}_{ij}$ = sample mean for the observations corresponding to the combination of treatment i (factor A) and treatment j (factor B)

$\bar{\bar{x}}$ = overall sample mean of all n_T observations

CAT Summary Data for The Two-factor Experiment

Factor A: Preparation Program	Factor B: College			Row totals	
	Business	Engineering	Arts and sciences		
Three-hour review	500 $\bar{x}_{11}=540$ 580	540 $\bar{x}_{12}= 500$ 460	480 $\bar{x}_{13}= 440$ 400	2960	
One-day program	460 $\bar{x}_{21}= 500$ 540	560 $\bar{x}_{22}= 590$ 620	420 $\bar{x}_{23}= 450$ 480	3080	
10-Week course	560 $\bar{x}_{31}= 580$ 600	600 $\bar{x}_{32}= 590$ 580	480 $\bar{x}_{33}= 445$ 410	3230	
Column totals	3240	3360	2670	Overall total= 9270	$\bar{x} = 515$

CAT Summary Data for The Two-factor Experiment

- Factor A means $\bar{x}_{1.} = 493.33$
 $\bar{x}_{2.} = 513.33$
 $\bar{x}_{3.} = 538.33$
- Factor B means $\bar{x}_{.1} = 540$
 $\bar{x}_{.2} = 560$
 $\bar{x}_{.3} = 445$

CAT Example:

Step 1. Compute the total sum of squares.

$$SST = \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^r (x_{ijk} - \bar{\bar{x}})^2$$

Step 1.
$$SST = (500 - 515)^2 + (580 - 515)^2 + (540 - 515)^2 + \cdots + (410 - 515)^2 = 82,450$$

CAT Example:

Step 2. Compute the sum of squares for factor A.

$$SSA = br \sum_{i=1}^a (\bar{x}_{i.} - \bar{\bar{x}})^2$$

Step 2. $SSA = (3)(2)[(493.33 - 515)^2 + (513.33 - 515)^2 + (538.33 - 515)^2] = 6100$

CAT Example:

Step 3. Compute the sum of squares for factor B.

$$SSB = ar \sum_{j=1}^b (\bar{x}_{.j} - \bar{\bar{x}})^2$$

Step 3. $SSB = (3)(2)[(540 - 515)^2 + (560 - 515)^2 + (445 - 515)^2] = 45,300$

CAT Example:

Step 4. Compute the sum of squares for interaction.

$$SSAB = r \sum_{i=1}^a \sum_{j=1}^b (\bar{x}_{ij} - \bar{x}_{i\cdot} - \bar{x}_{\cdot j} + \bar{\bar{x}})^2$$

Step 4. $SSAB = 2[(540 - 493.33 - 540 + 515)^2 + (500 - 493.33 - 560 + 515)^2 + \cdots + (445 - 538.33 - 445 + 515)^2] = 11,200$

CAT Example:

Step 5. Compute the sum of squares due to error.

$$SSE = SST - SSA - SSB - SSAB$$

Step 5. $SSE = 82,450 - 6100 - 45,300 - 11,200 = 19,850$

ANOVA Table for the **CAT** two-factor design

Sources of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F	P- value
Factor A	6100	2	3050	1.38	0.299
Factor B	45300	2	22650	10.27	0.005
Interaction	11200	4	2800	1.27	0.350
Error	19850	9	2206		
Total	82450	17			

Jupyter Code

```
In [15]: df2 = pd.read_excel('2way.xlsx')
```

```
In [16]: df2
```

Jupyter code

Out[16]:

	Value	prep_pro	college
0	500	three_hr	Business
1	580	three_hr	Business
2	540	three_hr	Engineering
3	460	three_hr	Engineering
4	480	three_hr	Artsandscience
5	400	three_hr	Artsandscience
6	460	One-day	Business
7	540	One-day	Business
8	560	One-day	Engineering
9	620	One-day	Engineering
10	420	One-day	Artsandscience
11	480	One-day	Artsandscience
12	560	10-Week	Business
13	600	10-Week	Business
14	600	10-Week	Engineering
15	580	10-Week	Engineering
16	480	10-Week	Artsandscience
17	410	10-Week	Artsandscience

Jupyter Code

```
In [20]: formula = 'Value ~C(college)+C(prepare)+C(college):C(prepare)'  
model = ols(formula, df2).fit()  
aov_table = anova_lm(model, typ=2)  
  
print(aov_table)
```

	sum_sq	df	F	PR(>F)
C(college)	45300.0	2.0	10.269521	0.004757
C(prepare)	6100.0	2.0	1.382872	0.299436
C(college):C(prepare)	11200.0	4.0	1.269521	0.350328
Residual	19850.0	9.0	NaN	NaN

Thank You





IIT ROORKEE



NPTEL ONLINE
CERTIFICATION COURSE

REGRESSION

Linear Regression

Dr. Ramesh Anbanandam

DEPARTMENT of Management Studies



Simple Linear Regression

- Simple Linear Regression Model
- Least Squares Method
- Coefficient of Determination
- Model Assumptions
- Testing for Significance
- Using the Estimated Regression Equation for Estimation and Prediction



Empirical Models

- Many problems in engineering and science involve exploring the relationships between two or more variables
- Regression analysis is a statistical technique that is very useful for these types of problems
- This model can also be used for process optimization, such as finding the level of temperature that maximizes yield, or for process control purposes

Empirical Models Example

- As an illustration, consider the data in the table.
- In this table y is the purity of oxygen produced in a chemical distillation process, and x is the percentage of hydrocarbons that are present in the main condenser of the distillation unit.

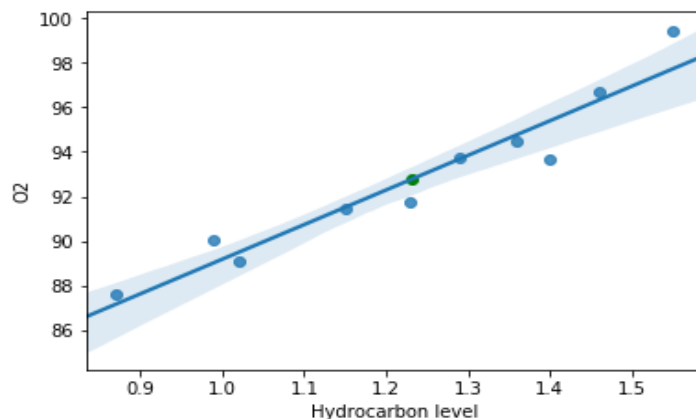
Hydrocarbon level (X)	Purity (Y)
0.99	90.01
1.02	89.05
1.15	91.43
1.29	93.74
1.46	96.73
1.36	94.45
0.87	87.59
1.23	91.77
1.55	99.42
1.4	93.65

Using python for plotting the data

```
In [20]: data = pd.read_excel('C:/Users/Somi/Desktop/reg2.xlsx')
```

```
In [19]: x= data['Hydrocarbon level']  
y = data['O2']  
plt.figure()  
sns.regplot(x,y,fit_reg= True)  
plt.scatter(np.mean(x), np.mean(y), color = "green")
```

```
Out[19]: <matplotlib.collections.PathCollection at 0x21ada0ab1d0>
```



Simple Linear Regression Model

- The equation that describes how y is related to x and an error term is called the regression model.
- The simple linear regression model is:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

where:

β_0 and β_1 are called parameters of the model,
 ε is a random variable called the error term.

Simple Linear Regression Equation

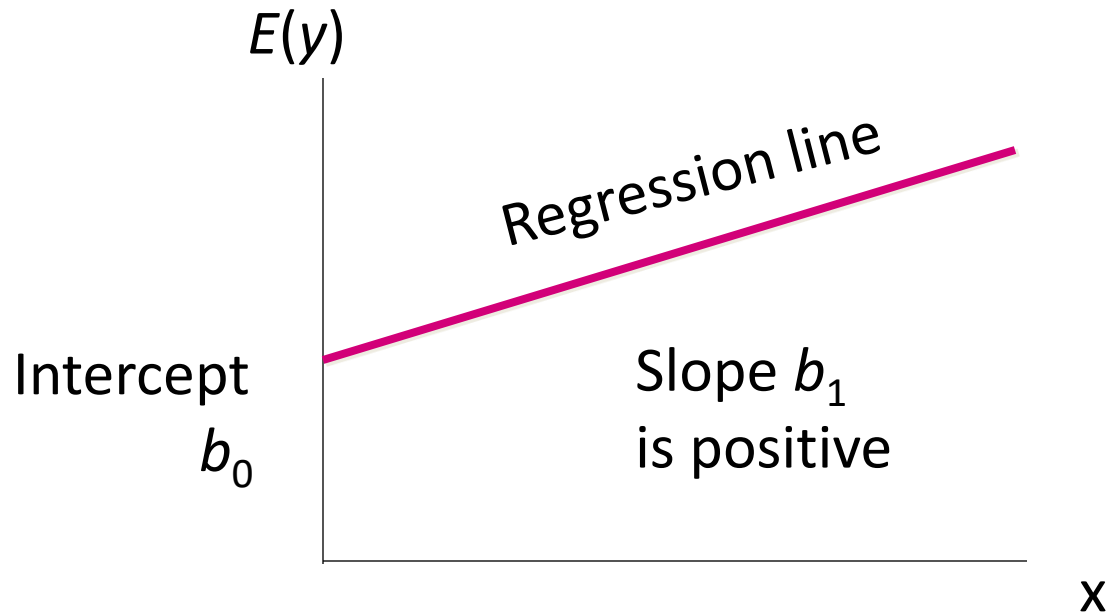
The simple linear regression equation is:

$$E(y) = \beta_0 + \beta_1 x$$

- Graph of the regression equation is a straight line.
- β_0 is the y intercept of the regression line.
- β_1 is the slope of the regression line.
- $E(y)$ is the expected value of y for a given x value.

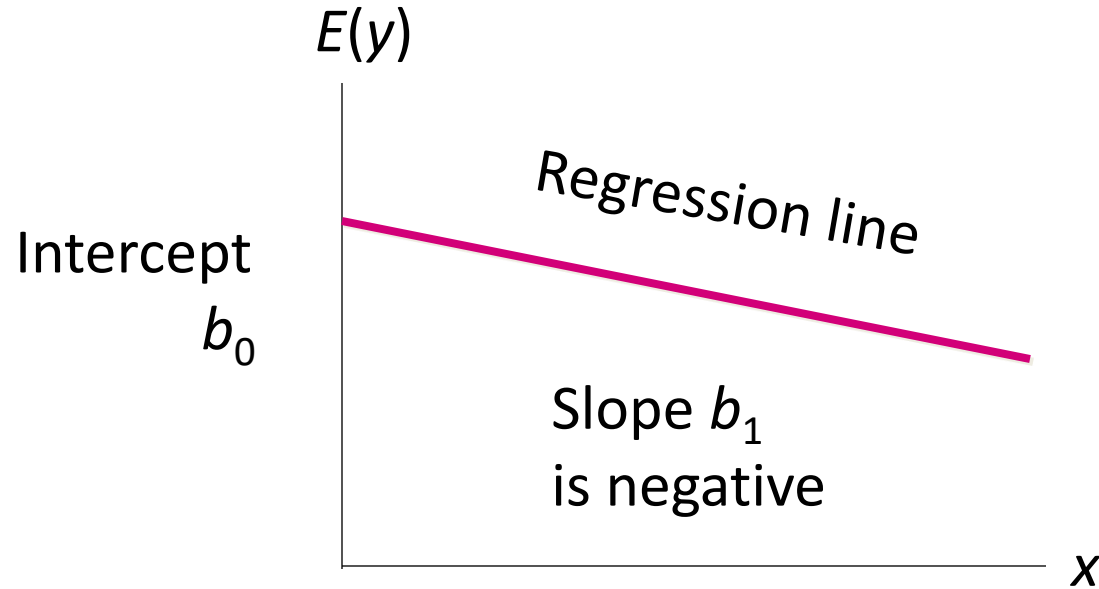
Simple Linear Regression Equation

Positive Linear Relationship



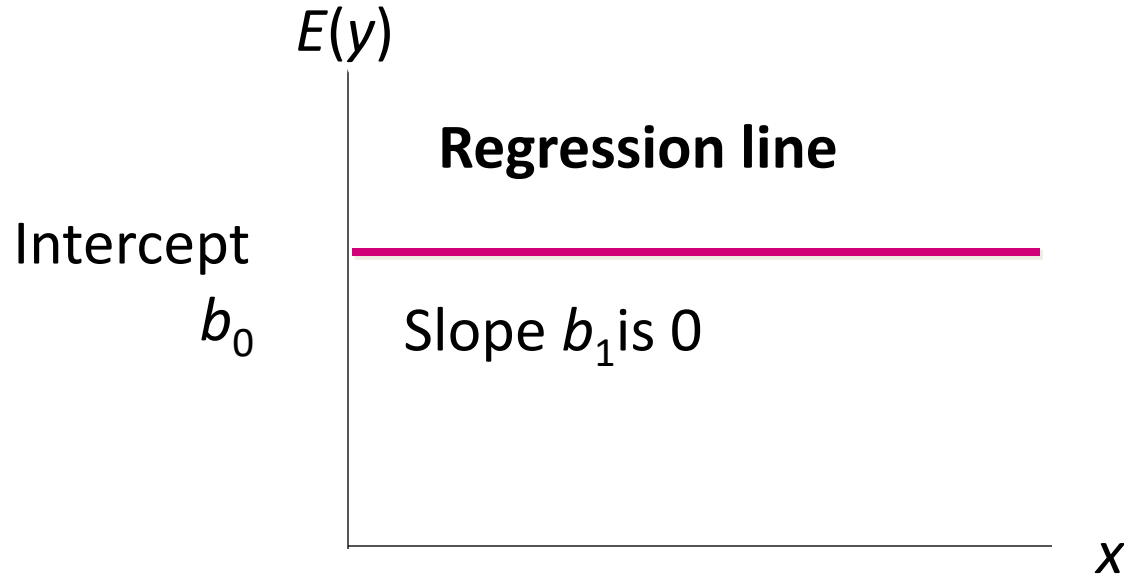
Simple Linear Regression Equation

Negative Linear Relationship



Simple Linear Regression Equation

No Relationship



Estimated Simple Linear Regression Equation

- The estimated simple linear regression equation

$$\hat{y} = b_0 + b_1x$$

- The graph is called the estimated regression line.
 - b_0 is the y intercept of the line.
 - b_1 is the slope of the line.
 - $\hat{y}$ is the estimated value of y for a given x value.

Least Squares Method

- Least Squares Criterion

$$\min \sum (y_i - \hat{y}_i)^2$$

where:

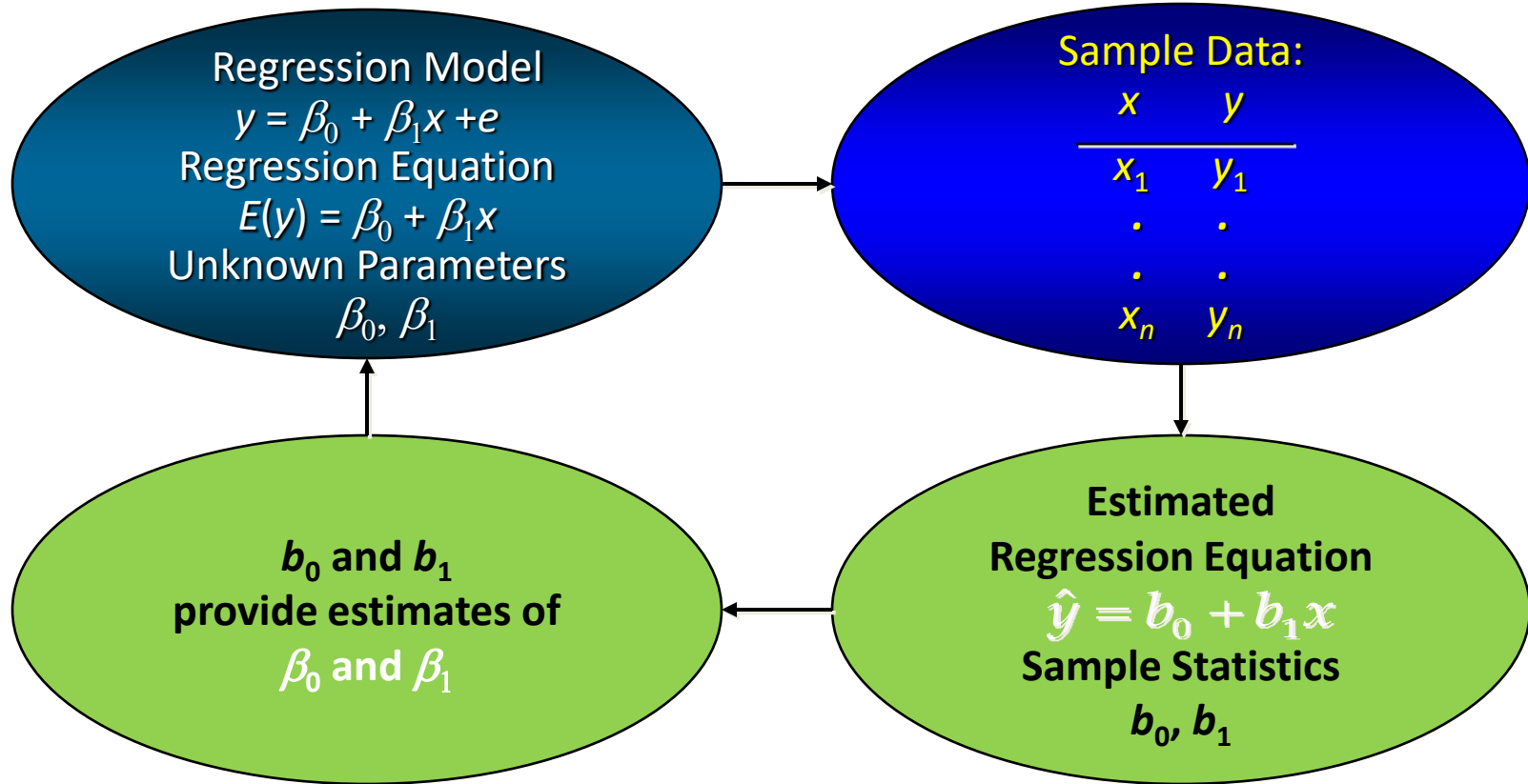
y_i = observed value of the dependent variable

for the i th observation

$\hat{y}_i$ = estimated value of the dependent variable

for the i th observation

Estimation Process





$$\begin{aligned}
 \text{Squared Error (SE)} &= (y_1 - (mx_1 + b))^2 + (y_2 - (mx_2 + b))^2 + \dots (y_n - (mx_n + b))^2 \\
 &= y_1^2 - 2y_1(mx_1 + b) + (mx_1 + b)^2 \\
 &\quad + y_2^2 - 2y_2(mx_2 + b) + (mx_2 + b)^2 \\
 &\quad + \dots \\
 &\quad + y_n^2 - 2y_n(mx_n + b) + (mx_n + b)^2 \\
 &= y_1^2 - 2x_1y_1m - 2y_1b + m^2x_1^2 + 2mx_1b + b^2 \\
 &\quad + y_2^2 - 2x_2y_2m - 2y_2b + m^2x_2^2 + 2mx_2b + b^2 \\
 &\quad + \dots \\
 &\quad + y_n^2 - 2x_ny_nm - 2y_nb + m^2x_n^2 + 2mx_nb + b^2
 \end{aligned}$$

$$\begin{aligned}
&= (y_1^2 + y_2^2 + \dots + y_n^2) \\
&\quad - 2m (x_1 y_1 + x_2 y_2 + \dots + x_n y_n) \\
&\quad - 2b (y_1 + y_2 + \dots + y_n) \\
&\quad + m^2 (x_1^2 + x_2^2 + \dots + x_n^2) \\
&\quad + 2mb (x_1 + x_2 + \dots + x_n) \\
&\quad + (b^2 + b^2 + \dots + b^2) \\
&= n \overline{y^2} - 2mn \overline{x y} - 2bn \overline{y} + m^2 n \overline{x^2} + 2mbn \overline{x} + nb^2
\end{aligned}$$

$$SE = n \overline{y}^2 - 2mn \overline{x} \overline{y} - 2bn \overline{y} + m^2 n \overline{x}^2 + 2mbn \overline{x} + nb^2$$

$$\frac{\partial(SE)}{\partial m} = -2n \overline{x} \overline{y} + 2mn \overline{x}^2 + 2bn \overline{x} = 0$$

$$\frac{\partial(SE)}{\partial m} = -2n \overline{x} \overline{y} + 2mn \overline{x}^2 + 2bn \overline{x} = 0$$

$$= -\overline{x} \overline{y} + m \overline{x}^2 + b \overline{x} = 0$$

$$m \overline{x}^2 + b \overline{x} = \overline{x} \overline{y}$$

$$m \frac{\overline{x}^2}{\overline{x}} + b = \frac{\overline{x} \overline{y}}{\overline{x}}$$

$$\text{one point } \left(\frac{\overline{x}^2}{\overline{x}}, \frac{\overline{x} \overline{y}}{\overline{x}} \right)$$

$$SE = n \bar{y}^2 - 2mn \bar{x} \bar{y} - 2bn \bar{y} + m^2 n \bar{x}^2 + 2mbn \bar{x} + nb^2$$

$$\frac{\partial(SE)}{\partial b} = -2n \bar{y} + 2mn \bar{x} + 2nb = 0$$

$$= -\bar{y} + m \bar{x} + b = 0$$

$$\bar{y} = m \bar{x} + b$$

another point $(\bar{x}, \bar{y})$







Least Squares Method

- Slope for the Estimated Regression Equation

$$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$



IIT ROORKEE



NPTEL ONLINE
CERTIFICATION COURSE

REGRESSION

Linear Regression-II

Dr. Ramesh Anbanandam

DEPARTMENT of Management Studies



Least Squares Method

- Slope for the Estimated Regression Equation

$$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

Sum of squares and sum of cross-products

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$$

$$S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x}) (y_i - \bar{y})$$

Sum of squares and sum of cross-products

$$\text{Slope}(m) = \frac{S_{xy}}{S_{xx}}$$

$$\text{SSE} = \text{error sum of squares} = S_{yy} - \frac{S_{xy}^2}{S_{xx}}$$

Least Squares Method

y-Intercept for the Estimated Regression Equation

$$b_0 = \bar{y} - b_1 \bar{x}$$

where:

x_i = value of independent variable for i th observation

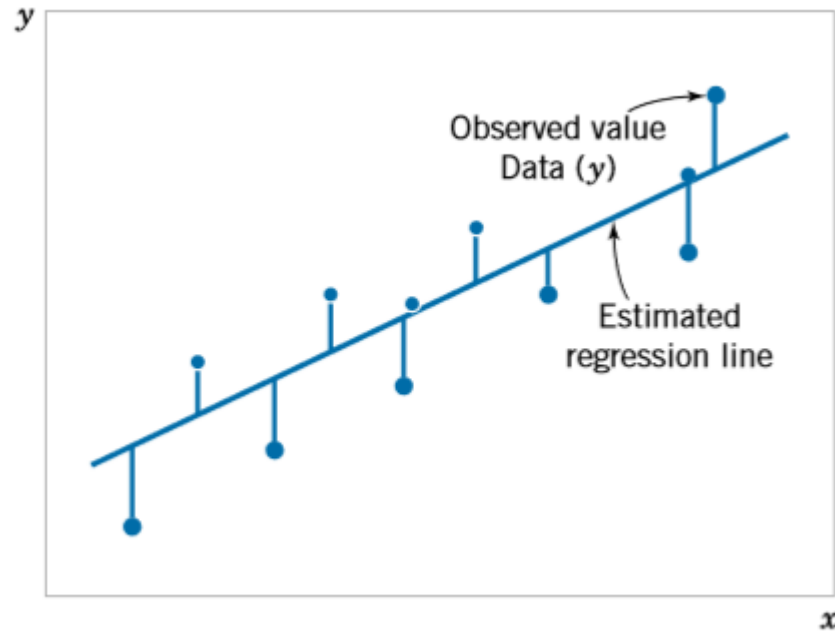
y_i = value of dependent variable for i th observation

$\bar{x}$ = mean value for independent variable

$\bar{y}$ = mean value for dependent variable

n = total number of observations

Simple Linear Regression



Deviation from the estimated regression model

Simple Linear Regression

Example: Auto Sales

An Auto company periodically has a special week-long sale.

As part of the advertising campaign runs one or more television commercials during the weekend preceding the sale.

Data from a sample of 5 previous sales are shown on the next slide.

Simple Linear Regression

Example: Auto Sales

<u>Number of TV Ads</u>	<u>Number of Cars Sold</u>
1	14
3	24
2	18
1	17
3	27

Estimated Regression Equation

Slope for the Estimated Regression Equation

$$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{20}{4} = 5$$

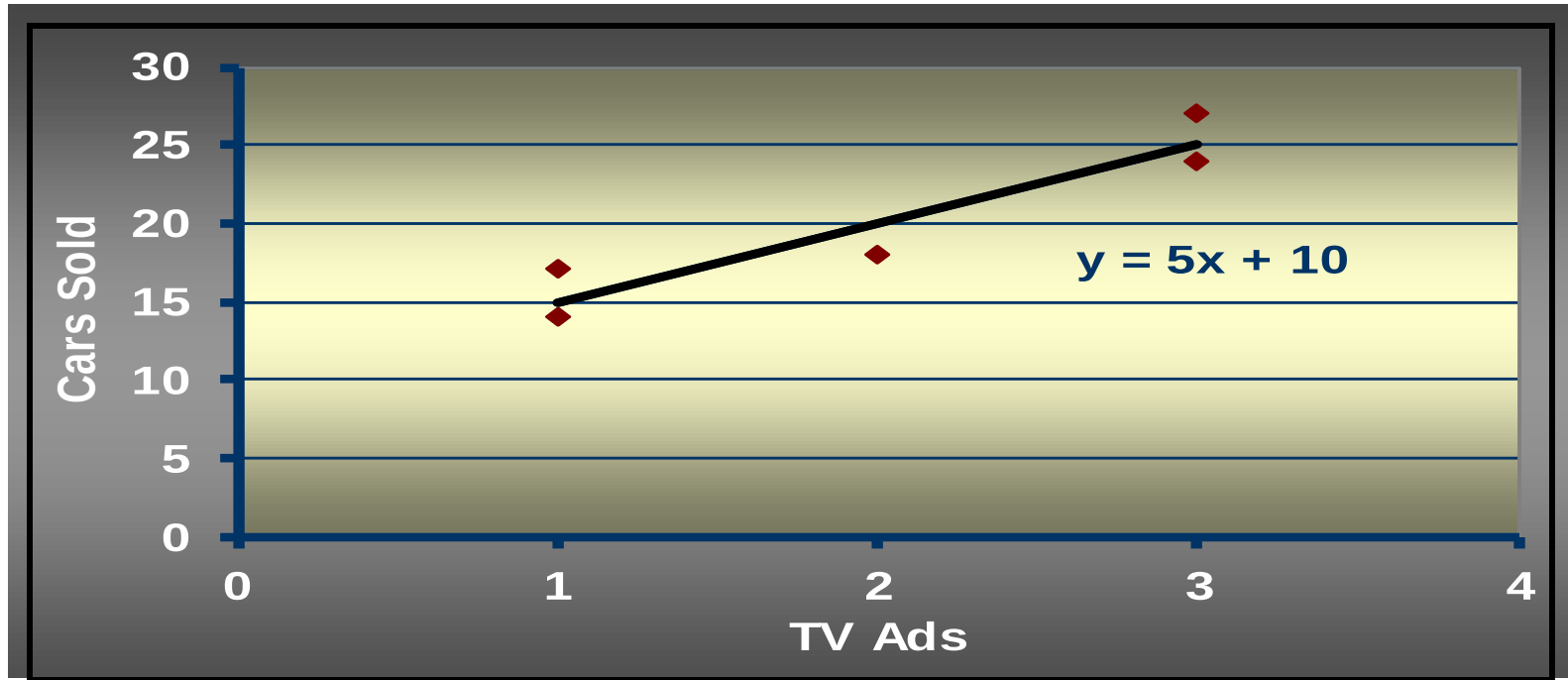
y-Intercept for the Estimated Regression Equation

$$b_0 = \bar{y} - b_1 \bar{x} = 20 - 5(2) = 10$$

Estimated Regression Equation

$$\hat{y} = 10 + 5x$$

Scatter Diagram and Trend Line



Jupyter Code

```
In [2]: import numpy as np  
import matplotlib.pyplot as plt
```

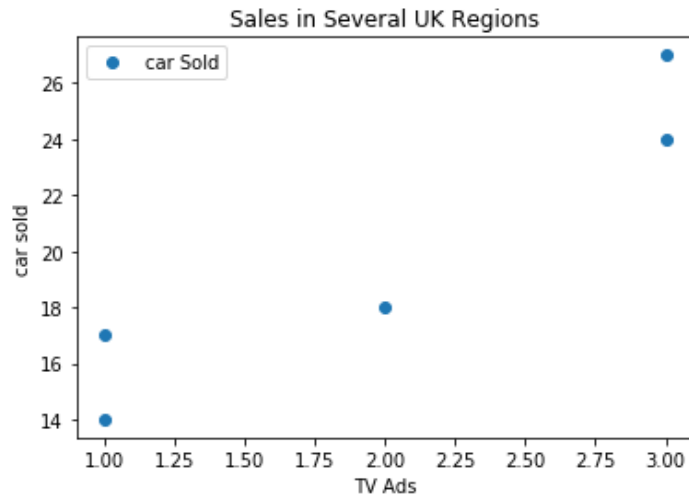
```
In [3]: import seaborn as sns
```

```
In [4]: import pandas as pd  
import matplotlib as mpl  
import statsmodels.formula.api as sm  
from sklearn.linear_model import LinearRegression  
from scipy import stats
```

```
In [5]: tbl = pd.read_excel('C:/Users/Somi/Documents/regr.xlsx')
```


Jupyter Code

```
In [6]: tbl.plot('TV Ads', 'car Sold', style='o')  
plt.ylabel('car sold')  
plt.title('Sales in Several UK Regions')  
plt.show()
```



Jupyter code

```
In [5]: t= tbl['TV Ads']  
c= tbl['car Sold']
```

```
In [8]: import statsmodels.api as sm  
t = sm.add_constant(t)  
model1 = sm.OLS(c,t)  
result1 = model1.fit()  
print(result1.summary())
```

```
OLS Regression Results  
=====
```

Dep. Variable:	car Sold	R-squared:	0.877
Model:	OLS	Adj. R-squared:	0.836
Method:	Least Squares	F-statistic:	21.43
Date:	Fri, 30 Aug 2019	Prob (F-statistic):	0.0190
Time:	08:31:20	Log-Likelihood:	-9.6687
No. Observations:	5	AIC:	23.34
Df Residuals:	3	BIC:	22.56
Df Model:	1		
Covariance Type:	nonrobust		

```
=====
```

	coef	std err	t	P> t	[0.025	0.975]
const	10.0000	2.366	4.226	0.024	2.469	17.531
TV Ads	5.0000	1.080	4.629	0.019	1.563	8.437

```
=====
```

Omnibus:	nan	Durbin-Watson:	1.214
Prob(Omnibus):	nan	Jarque-Bera (JB):	0.674
Skew:	0.256	Prob(JB):	0.714
Kurtosis:	1.276	Cond. No.	6.33

```
=====
```

Example Problem- II

- The data in the file hardness.xls provide measurements on the hardness and tensile strength for 35 specimens of die-cast aluminum.
 - It is believed that hardness (measured in Rockwell E units) can be used to predict tensile strength (measured in thousands of pounds per square inch).
- a. Construct a scatter plot.
 - b. Assuming a linear relationship, use the least-squares method to find the regression coefficients b_0 and b_1 .
 - c. Interpret the meaning of the slope, b_1 , in this problem.
 - d. Predict the mean tensile strength for die-cast aluminum that has a hardness of 30 Rockwell E units.

Tensile strength	Hardness
53	29.31
70.2	34.86
84.3	36.82
55.3	30.12
78.5	34.02
63.5	30.82
71.4	35.4
53.4	31.26
82.5	32.18
67.3	33.42
69.5	37.69
73	34.88
55.7	24.66
85.8	34.76
95.4	38.02
51.1	25.68
74.4	25.81
54.1	26.46
77.8	28.67
52.4	24.64
69.1	25.77
53.5	23.69
64.3	28.65
82.7	32.38
55.7	23.21
70.5	34
87.5	34.47
50.7	29.25
72.3	28.71
59.5	29.83
71.3	29.25
52.7	27.99
76.5	31.85
63.7	27.65
69.2	31.7

```
In [30]: ► mean_squared_error(y_test, y_predict)
```

```
Out[30]: 35.71053398209997
```

```
In [31]: ► reg.score(x_test, y_test)
```

```
Out[31]: 0.5362243730094254
```

```
In [32]: ► reg.score(x_train, y_train)
```

```
Out[32]: 0.4500146647765303
```

Thank You





IIT ROORKEE



NPTEL ONLINE
CERTIFICATION COURSE

REGRESSION

Linear Regression-III

Dr. Ramesh Anbanandam

DEPARTMENT of Management Studies



Learning Objectives

- Understanding Coefficient of Determination
- Test statistical hypotheses and construct confidence intervals on regression model parameters



Coefficient of Determination

- Relationship Among SST, SSR, SSE

$$SST = SSR + SSE$$


$$\sum (y_i - \bar{y})^2 = \sum (\hat{y}_i - \bar{y})^2 + \sum (y_i - \hat{y}_i)^2$$

$$SS_{yy} = \left(\frac{SS_{xy}^2}{SS_{xx}} \right) + \left(SS_{yy} - \frac{SS_{xy}^2}{SS_{xx}} \right)$$

where:

SST = total sum of squares

SSR = sum of squares due to regression

SSE = sum of squares due to error

Coefficient of Determination

- The coefficient of determination is:

$$r^2 = SSR/SST$$

where:

SSR = sum of squares due to regression

SST = total sum of squares

Coefficient of Determination

$$r^2 = SSR/SST = 100/114 = .8772$$

The regression relationship is very strong; 88% of the variability in the number of cars sold can be explained by the linear relationship between the number of TV ads and the number of cars sold.



Jupyter code

```
In [5]: t= tbl['TV Ads']  
c= tbl['car Sold']
```

```
In [8]: import statsmodels.api as s  
t = s.add_constant(t)  
model1 = sm.OLS(c,t)  
result1 = model1.fit()  
print(result1.summary())
```

OLS Regression Results

Dep. Variable:	car Sold	R-squared:	0.877
Model:	OLS	Adj. R-squared:	0.836
Method:	Least Squares	F-statistic:	21.43
Date:	Fri, 30 Aug 2019	Prob (F-statistic):	0.0190
Time:	08:31:20	Log-Likelihood:	-9.6687
No. Observations:	5	AIC:	23.34
Df Residuals:	3	BIC:	22.56
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	10.0000	2.366	4.226	0.024	2.469	17.531
TV Ads	5.0000	1.080	4.629	0.019	1.563	8.437

Omnibus:	nan	Durbin-Watson:	1.214
Prob(Omnibus):	nan	Jarque-Bera (JB):	0.674
Skew:	0.256	Prob(JB):	0.714
Kurtosis:	1.276	Cond. No.	6.33

Coefficient of Determination

$$r^2 = SSR/SST = 100/114 = .8772$$

The regression relationship is very strong; 88% of the variability in the number of cars sold can be explained by the linear relationship between the number of TV ads and the number of cars sold.

Sample Correlation Coefficient

$$r_{xy} = (\text{sign of } b_1) \sqrt{\text{Coefficient of Determination}}$$

$$r_{xy} = (\text{sign of } b_1) \sqrt{r^2}$$

$$\hat{y} = b_0 + b_1x$$

where:

b_1 = the slope of the estimated regression equation

Sample Correlation Coefficient

$$r_{xy} = (\text{sign of } b_1) \sqrt{r^2}$$

The sign of b_1 in the equation $\hat{y} = 10 + 5x$ is “+”.

$$r_{xy} = +\sqrt{.8772}$$

$$r_{xy} = +.9366$$

Assumptions About the Error Term e

1. The error e is a random variable with mean of zero.
2. The variance of e , denoted by e^2 , is the same for all values of the independent variable.
3. The values of e are independent.
4. The error e is a normally distributed random variable.



Testing for Significance

- To test for a significant regression relationship, we must conduct a hypothesis test to determine whether the value of β_1 is zero.
- Two tests are commonly used:

t Test

and

F Test

- Both the *t* test and *F* test require an estimate of s^2 , the variance of e in the regression model.

Estimate of s

- An Estimate of s

The mean square error (MSE) provides the estimate of s^2 , and the notation s^2 is also used.

$$s^2 = \text{MSE} = \text{SSE}/(n - 2)$$

where:

$$\text{SSE} = \sum (y_i - \hat{y}_i)^2 = \sum (y_i - b_0 - b_1 x_i)^2$$

Testing for Significance

- An Estimate of s
 - To estimate s we take the square root of s^2 .
 - The resulting s is called the standard error of the estimate.

$$s = \sqrt{\text{MSE}} = \sqrt{\frac{\text{SSE}}{n-2}}$$

Testing for Significance

S_e = Standard error of the estimate (σ^2)

$$= \frac{SSE}{n-2} = \frac{S_{yy} - \frac{S_{xy}^2}{S_{xx}}}{n-2}$$

Testing for Significance: t Test

- Hypotheses

$$H_0: \beta_1 = 0$$

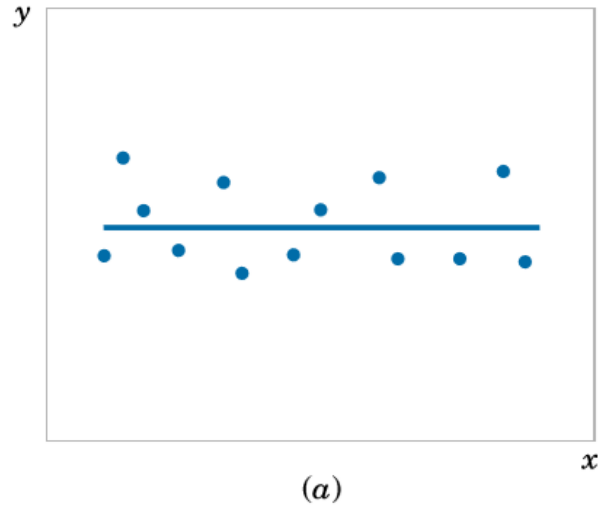
$$H_a: \beta_1 \neq 0$$

- Test Statistic

$$t = \frac{b_1}{s_{b_1}}$$

Case 1

$$H_0: \beta_1 = 0$$

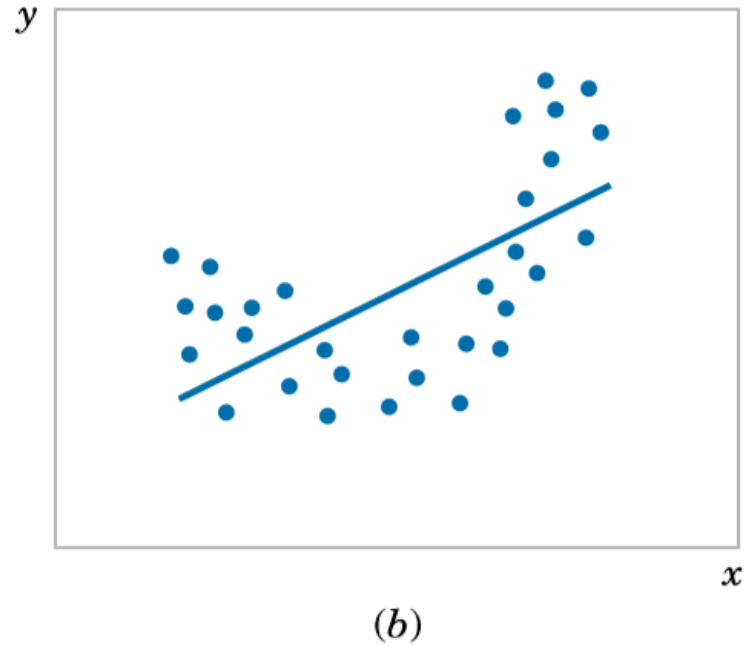


In this case hypothesis is not rejected

Case 2

$$H_a: \beta_1 \neq 0$$

In this case hypothesis is rejected



The Standard Deviation of the Regression Slope

- The standard error of the regression slope coefficient (b_1) is estimated by

$$s_{b_1} = \frac{s_\varepsilon}{\sqrt{\sum (x - \bar{x})^2}} = \frac{s_\varepsilon}{\sqrt{\sum x^2 - \frac{(\sum x)^2}{n}}}$$

where:

s_{b_1} = Estimate of the standard error of the least squares slope

$s_\varepsilon = \sqrt{\frac{SSE}{n-2}}$ = Sample standard error of the estimate

Testing for Significance: t Test

■ Rejection Rule

Reject H_0 if $p\text{-value} \leq \alpha$
or $t \leq -t_{\alpha/2}$ or $t \geq t_{\alpha/2}$

where:

$t_{\alpha/2}$ is based on a t distribution
with $n - 2$ degrees of freedom

Testing for Significance: t Test

1. Determine the hypotheses.

$$H_0: \beta_1 = 0$$

$$H_a: \beta_1 \neq 0$$

2. Specify the level of significance. $\alpha = .05$

3. Select the test statistic.

$$t = \frac{b_1}{s_{b_1}}$$

4. State the rejection rule.

Reject H_0 if $p\text{-value} \leq .05$
or $|t| > 3.182$ (with
3 degrees of freedom)

Testing for Significance: t Test

5. Compute the value of the test statistic.

$$t = \frac{b_1 - \beta_1}{s_{b_1}} = \frac{5}{1.08} = 4.63$$

6. Determine whether to reject H_0 .

$t = 4.541$ provides an area of .01 in the upper tail. Hence, the p -value is less than .02. (Also, $t = 4.63 > 3.182$.) We can reject H_0 .

Hypothesis Tests for the Slope of the Regression Model

$$H_0: \beta_1 = 0$$

$$H_1: \beta_1 \neq 0$$

$$H_0: \beta_1 \leq 0$$

$$H_1: \beta_1 > 0$$

$$H_0: \beta_1 \geq 0$$

$$H_1: \beta_1 < 0$$

$$t = \frac{b_1 - \beta_1}{S_b}$$

where: $S_b = \frac{S_e}{\sqrt{SS_{XX}}}$

$$S_e = \sqrt{\frac{SSE}{n-2}}$$

$$SS_{XX} = \sum X^2 - \frac{(\sum X)^2}{n}$$

β_1 = the hypothesized slope

$$df = n - 2$$

Confidence Interval for β_1

- We can use a 95% confidence interval for β_1 to test the hypotheses just used in the t test.
- H_0 is rejected if the hypothesized value of β_1 is not included in the confidence interval for β_1 .

Confidence Interval for β_1

- The form of a confidence interval for β_1 is:

$$b_1 \pm t_{\alpha/2} s_{b_1}$$

b_1 is the
point
estimator

$t_{\alpha/2} s_{b_1}$
is the
margin
of error

Where $t_{\alpha/2}$ is the t value providing an area of $\alpha/2$ in the upper tail of a t distribution with $n - 2$ degrees of freedom

Confidence Interval for β_1

- Rejection Rule

Reject H_0 if 0 is not included in the confidence interval for β_1 .

- 95% Confidence Interval for β_1

$$b_1 \pm t_{\alpha/2} s_{b_1} = 5 \pm 3.182(1.08) = 5 \pm 3.44$$

or 1.56 to 8.44

- Conclusion

0 is not included in the confidence interval.

Reject H_0

Testing for Significance: F Test

- Hypotheses

$$H_0: \beta_1 = 0$$

$$H_a: \beta_1 \neq 0$$

- Test Statistic

$$F = MSR/MSE$$

F-Test for Significance

- F Test statistic:

$$F = \frac{MSR}{MSE}$$

where

$$MSR = \frac{SSR}{k}$$
$$MSE = \frac{SSE}{n - k - 1}$$

where F follows an F distribution with k numerator degrees of freedom and (n - k - 1) denominator degrees of freedom
(k = the number of independent variables in the regression model)

Testing for Significance: F Test

- Rejection Rule

Reject H_0 if
 $p\text{-value} \leq \alpha$
or $F \geq F_\alpha$

where:

F_α is based on an F distribution with
1 degree of freedom in the numerator and
 $n - 2$ degrees of freedom in the denominator

Testing for Significance: F Test

1. Determine the hypotheses.

$$H_0: \beta_1 = 0$$

$$H_a: \beta_1 \neq 0$$

2. Specify the level of significance. $\alpha = .05$

3. Select the test statistic. $F = \text{MSR}/\text{MSE}$

4. State the rejection rule. Reject H_0 if $p\text{-value} \leq .05$
or $F \geq 10.13$ (with 1 d.f.
in numerator and
3 d.f. in denominator)



Jupyter Code

```
In [2]: import numpy as np  
import matplotlib.pyplot as plt
```

```
In [3]: import seaborn as sns
```

```
In [4]: import pandas as pd  
import matplotlib as mpl  
import statsmodels.formula.api as sm  
from sklearn.linear_model import LinearRegression  
from scipy import stats
```

```
In [5]: tbl = pd.read_excel('C:/Users/Somi/Documents/regr.xlsx')
```

Jupyter code

```
In [5]: t= tbl['TV Ads']  
c= tbl['car Sold']
```

```
In [8]: import statsmodels.api as sm  
t = sm.add_constant(t)  
model1 = sm.OLS(c,t)  
result1 = model1.fit()  
print(result1.summary())
```

```
OLS Regression Results  
=====
```

Dep. Variable:	car Sold	R-squared:	0.877
Model:	OLS	Adj. R-squared:	0.836
Method:	Least Squares	F-statistic:	21.43
Date:	Fri, 30 Aug 2019	Prob (F-statistic):	0.0190
Time:	08:31:20	Log-Likelihood:	-9.6687
No. Observations:	5	AIC:	23.34
Df Residuals:	3	BIC:	22.56
Df Model:	1		
Covariance Type:	nonrobust		

```
=====
```

	coef	std err	t	P> t	[0.025	0.975]
const	10.0000	2.366	4.226	0.024	2.469	17.531
TV Ads	5.0000	1.080	4.629	0.019	1.563	8.437

```
=====
```

Omnibus:	nan	Durbin-Watson:	1.214
Prob(Omnibus):	nan	Jarque-Bera (JB):	0.674
Skew:	0.256	Prob(JB):	0.714
Kurtosis:	1.276	Cond. No.	6.33

```
=====
```

Testing for Significance: F Test

5. Compute the value of the test statistic.

$$F = \text{MSR}/\text{MSE} = 100/4.667 = 21.43$$

6. Determine whether to reject H_0 .

$F = 17.44$ provides an area of .025 in the upper tail. Thus, the p -value corresponding to $F = 21.43$ is less than $2(.025) = .05$. Hence, we reject H_0 .

The statistical evidence is sufficient to conclude that we have a significant relationship between the number of TV ads aired and the number of cars sold.



Some Cautions about the Interpretation of Significance Tests

- Rejecting $H_0: \beta_1 = 0$ and concluding that the relationship between x and y is significant does not enable us to conclude that a cause-and-effect relationship is present between x and y .
- Just because we are able to reject $H_0: \beta_1 = 0$ and demonstrate statistical significance does not enable us to conclude that there is a linear relationship between x and y .



Thank You

